Week 5 Assignment: Deep Learning with CNN on Fashion-MNIST

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Deep Learning Model Training and Evaluation using HuggingFace Datasets and Keras

Dataset & Preprocessing

The Fashion-MNIST dataset (10 clothing categories, 28×28 grayscale images) was loaded from the HuggingFace Datasets library. The predefined training split was further divided into 90% training and 10% validation subsets to support hyperparameter tuning and early stopping. The official test split was kept completely untouched and used only for final evaluation to ensure unbiased performance estimation. Pixel values were normalized to the range $[0, 1]$, and images were reshaped to include a single channel dimension (28,28,1).

Model Architecture

A Convolutional Neural Network (CNN) was implemented to capture spatial patterns in image data. The architecture consisted of:

- Conv2D(32 filters, 3×3 kernel) + ReLU activation
- MaxPooling layer
- Conv2D(64 filters, 3×3 kernel) + ReLU activation
- MaxPooling layer
- Flatten layer
- Dense(128 units, ReLU)
- Dropout(0.3) for regularization
- Dense(10 units, Softmax output)

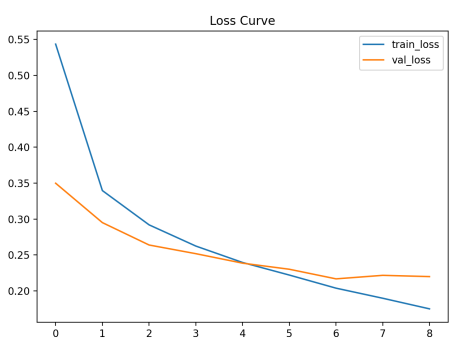
The model was trained using the Adam optimizer with sparse categorical cross-entropy loss. Early stopping monitored validation loss to prevent overfitting.

Evaluation Results

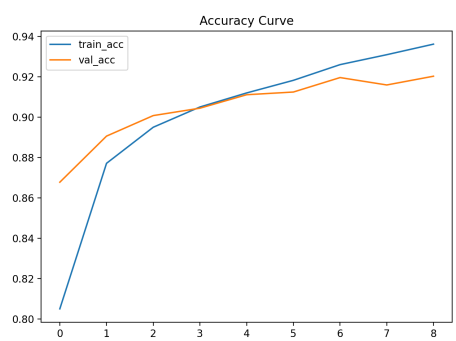
- Test Accuracy: **0.9122**
- Key aggregate metrics:

```
accuracy                           0.9122     10000
macro avg      0.9118    0.9122    0.9115     10000
weighted avg   0.9118    0.9122    0.9115     10000
```

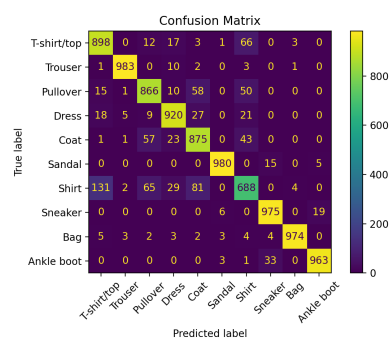
- Most frequent confusions:
  - Shirt → T-shirt/top: 131 cases
  - Shirt → Coat: 81 cases
  - T-shirt/top → Shirt: 66 cases



Loss Curve



Accuracy Curve



Confusion Matrix

Discussion and Insights

The CNN effectively learned hierarchical spatial features such as edges and texture patterns, achieving over 91% test accuracy. The confusion matrix reveals that most errors occur between visually similar garment categories, particularly Shirt and T-shirt/top, indicating overlapping shape and texture features. Training and validation curves demonstrate stable convergence with limited overfitting, confirming that dropout and early stopping contributed to generalization performance.